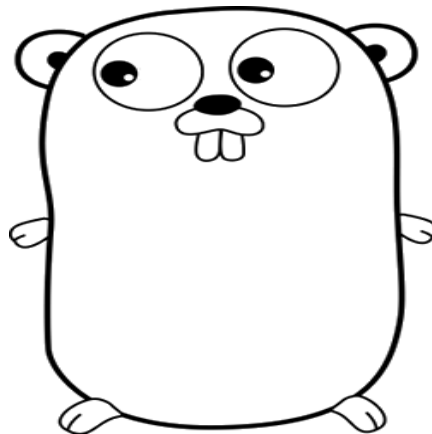


Nautilus PDF Library

Complete Developer Feature Demo

v1.0 • Go 1.26+

Generated: 23 March 2026



What this demo covers

- Low-level PDF drawing: text, lines, rectangles, polygons, circles
- Border patterns: solid, dashed, dotted, dash-dot
- Frames with backgrounds, padding, and borders
- Tables with colspan / rowspan, per-cell styling, page overflow
- Images (PNG Gopher artwork)
- Emoji (inline PNG via Noto Emoji resolver)
- Right-to-left text: Arabic shaping + Hebrew BiDi
- Layout engine: DocTemplate, PageTemplate, Flowables
- All 20 Highcharts-style chart types



Typography & Text Rendering

Font Sizes

8pt — The quick brown fox jumps over the lazy gopher

9pt — The quick brown fox jumps over the lazy gopher

10pt — The quick brown fox jumps over the lazy gopher

11pt — The quick brown fox jumps over the lazy gopher

12pt — The quick brown fox jumps over the lazy gopher

14pt — The quick brown fox jumps over the lazy gopher

16pt — The quick brown fox jumps over the lazy gopher

18pt — The quick brown fox jumps over the lazy gopher

20pt — Go makes it simple

24pt — Go makes it simple

28pt — Go makes it simple

32pt — Go makes it simple

36pt — Go makes it simple

Text Colors

Navy (0,38,84) — the Go Gopher says hello!

Cyan (0,172,215) — the Go Gopher says hello!

Coral (220,80,60) — the Go Gopher says hello!

Mint (0,164,130) — the Go Gopher says hello!

Gold (255,193,7) — the Go Gopher says hello!

Lavender (140,100,200) — the Go Gopher says hello!

Slate (90,103,120) — the Go Gopher says hello!

Standard Black (0,0,0) — the Go Gopher says hello!

Word-Wrapped Body Text (Go-themed)

Go is an open source programming language that makes it simple to build secure, scalable software. Developed at Google, Go combines the simplicity of scripting with the performance of compiled languages. The Go runtime includes garbage collection, goroutines, and channels — primitives that make concurrent programming straightforward and powerful.

The Go Gopher was designed by Renee French and first appeared in 2009 when Go was first announced. The Gopher has become the beloved mascot of the Go community, appearing in countless community contributions, conference talks, and merchandise.

Emoji — Inline PNG Rendering

No emoji directory provided. Run with `-emoji assets/emoji/png/128` to enable.

Right-to-Left Text — Arabic & Hebrew

Nautilus supports right-to-left scripts via the pdf/rtl package. Arabic text is automatically shaped (contextual letter forms + lam-alef ligatures) and reordered using the Unicode Bidirectional Algorithm. Hebrew uses BiDi reordering without additional shaping. Both scripts render correctly in frames, inline mixed-direction text, and multi-line word-wrapped paragraphs.

Arabic — rtl.Shape() + WriteLineRTL / WriteTextRTL

(kitāb — book)

كتا .

(madrasa — school)

مدرسة

(hello world)

مرحبا بالعالم

(shukran jazīlan — thank you very much)

شكرا جزيلا

lam-alef ligatures:

لا لآ لا لآ

mixed number — 42

دولار السعر

mixed Latin brand — Google

للتكنولوجيا شركة

mixed punctuation — !

هل أنت مستعد؟ نعم!

Prose paragraph — (WriteTextRTL, font size 15):

في عالم يتسارع فيه التطور التكنولوجي بشكل لم يسبق له مثيل، أصبح إتقان دوا البرمجة للمطورين القدر الحديثة ضرور لا غنى عنها لكل مطور يسعى إلى النجاح. تقدم مكتبة دوا الحاجة إلى تبعية خارجية عالية الجودة مباشرة من لغة على توليد ملفا معقد.

يدعم النظام اللغة العربية بشكل كامل، بما في ذلك التشكيل السياقي للحروف، وربط اللام ثنائية الاتجاه. كما يمكن بالألف في جميع حالاته، وإعادة الترتيب البصري وفق خوارزمية دمج النصوص العربية مع النصوص اللاتينية والأرقام في نفس السطر بسلاسة تامة.

Arabic proverb inside a Frame (right-side gold accent border):

من طلب العلا سهر الليالي

(Arabic proverb: "He who seeks glory stays awake at night")

Longer Arabic text block inside a Frame:

. تتميز هذ المكتبة بدعمها باستخدام لغة توليد ملفا مرحبا بكم في مكتبة
الكامل للخطوط العربية، وإمكانية الجمع بين النصوص ذا الاتجاهات المختلفة في
صفحة واحد. يمكنك إنشاء جداول، ورسوم بيانية، وتخطيطا معقد متعدد الأعمد
بكل سهولة.

Arabic vocabulary table – three columns		
Arabic	Transliteration	English
بيت	bayt	house
ماء	mā'	water
شمس	shams	sun
قمر	qamar	moon
نور	nūr	light
حيا	ḥayāt	life

Right-to-Left Text — Hebrew

Hebrew is written right-to-left and shares the Unicode Bidirectional Algorithm with Arabic. Unlike Arabic, Hebrew letters do not change shape depending on their position in a word, so no contextual shaping is required — `rtl.Shape()` passes the text unchanged. The `pdf/rtl` package applies BiDi reordering so that glyphs are placed in visual (right-to-left) order on the page.

Hebrew — basic vocabulary (WriteLineRTL)

(shalom — peace / hello / goodbye)

שלום

(todah — thank you)

תודה

(todah rabah — thank you very much)

תודה רבה

(bevakasha — please / you're welcome)

בבקשה

/ (ken / lo — yes / no)

כן / לא

(Yisra'el — Israel)

ישראל

(Yerushalayim — Jerusalem)

ירושלים

(Shabbat Shalom — Sabbath greeting)

שבת שלום

mixed — Google

הוא מנוע חיפוש פופולרי

mixed number — 99

שקלים המחיר הוא

Biblical Hebrew — Genesis 1:1-3 (WriteTextRTL)

בְּרֵאשִׁית בָּרָא אֱלֹהִים אֶת הַשָּׁמַיִם וְאֶת הָאָרֶץ
וְהָאָרֶץ הָיְתָה תֹהוֹ וָבֹהוּ וְחָשֶׁךְ עַל פְּנֵי תְהוֹם וְרוּחַ אֱלֹהִים מְרַחֶפֶת עַל פְּנֵי הַמַּיִם
וַיֹּאמֶר אֱלֹהִים יְהי אוֹר וַיְהי אוֹר

Genesis 1:1-3 (Hebrew Bible, public domain)

Modern Hebrew prose – technology paragraph (WriteTextRTL)

בעידן הדיגיטלי המודרני פיתוח תוכנה הפך לאחד המקצועות המבוקשים ביותר בשוק זכתה לפופולריות רבה בשנת שפותחה על ידי העבודה הגלובלי. שפת התכנות בקרב מפתחים בשל פשטותה ביצועיה הגבוהים ותמיכתה המובנית בתכנות מקבילי.

מורכבים ישירות מהקוד ללא תלות ליצור מסמכי מאפשרת למפתחי ספריית בספריות חיצוניות. הספרייה תומכת בעברית בערבית ובשפות נוספות הנכתבות מימין הדו כיווני לסידור הנכון של הטקסט. לשמאל תוך שימוש באלגוריתם

בנוסף ניתן לשלב טקסט עברי עם מילים בלטינית באותו משפט כפי שמוצג בדוגמאות לעיל. יכולת זו חיונית ביישומים עסקיים רבים כגון חשבוניות דוחות כספיים ומסמכים רשמיים.

Hebrew proverb inside a Frame

כָּל יִשְׂרָאֵל יֵשׁ לָהֶם חֵלֶק לְעוֹלָם הַבָּא

(Sanhedrin 10:1 – "All Israel have a share in the world to come")

דוגמה זו מציגה טקסט עברי ארוך יותר בתוך מסגרת. המסגרת מספקת רקע וגבולות ומאפשרת ארגון ויזואלי ברור של התוכן. הטקסט נאסף באופן אוטומטי בתוך גבולות המסגרת.

Side-by-side: LTR (English) | RTL (Arabic) | RTL (Hebrew)**English (LTR)**

The quick brown fox jumps over
the lazy dog. Programming in Go
is fast and fun.

Arabic (RTL)

الثعلب البني السريع يقفز فوق
الكلب الكسول. البرمجة بلغة
سريعة وممتعة.

Hebrew (RTL)

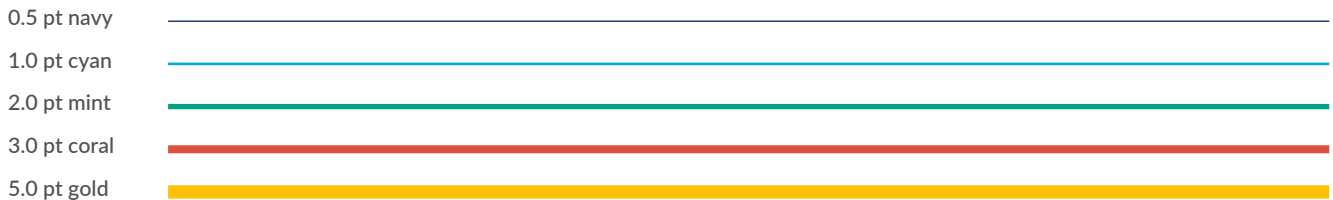
השועל החום המהיר קופץ
מעל הכלב העצלן. תכנות
מהיר ומהנה. ב

Drawing Primitives

Filled Rectangles



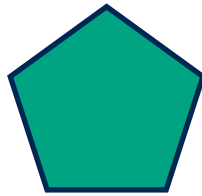
Lines — Various Widths & Colors



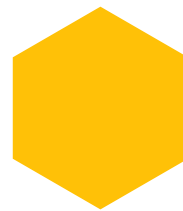
Polygons



Filled triangle

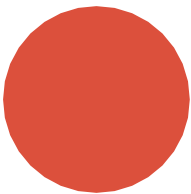


Filled+stroked pentagon

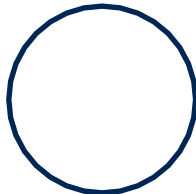


Filled hexagon

Circles



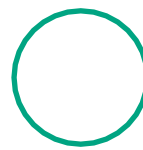
Filled



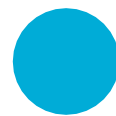
Stroked 1pt



Filled



Stroked 2pt



Filled

Graphics State: SaveGraphicsState / RestoreGraphicsState

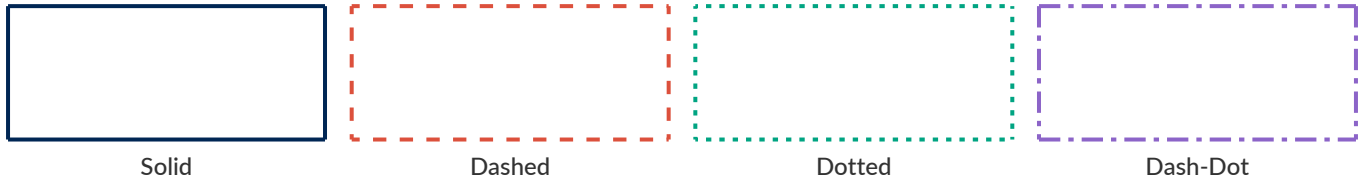


Three rects drawn with nested Save/Restore state

Go's standard library is comprehensive: it includes packages for HTTP, cryptography, JSON, SQL, testing, and much more. The toolchain ships as a single binary and supports cross-compilation to dozens of operating system and architecture combinations with a single command.

Borders & Frames

Border Patterns (all 4 built-in styles)



Solid

Dashed

Dotted

Dash-Dot

Selective Sides (top only, bottom only, left+right)

Top only

Bottom only (dashed)

Left+Right (dotted)



Frames: auto-height, fixed-height with background, padding

Auto-height frame with a solid navy border and 8 pt uniform padding. Content flows down and the border adapts.

Fixed 90 pt height, light-blue background, 1.5 pt cyan border.

Dark navy background, white text, no explicit border, 10 pt padding on all sides.

Two-column frame layout using pdf.Frame

Left column: Goroutines are lightweight threads managed by the Go runtime. Unlike OS threads, goroutines start with just a few kilobytes of stack space that can grow and shrink on demand. You can launch thousands of goroutines without exhausting system memory.

Right column: Go modules, introduced in Go 1.11 and made the default in Go 1.16, provide a robust dependency management system. Each module has a go.mod file that declares its dependencies with exact version pinning, ensuring reproducible builds across machines and time.

`doc.DrawBorder` draws per-side borders directly on the page. `doc.NewFrame` wraps content with optional background, padding, and border, calling `doc.DrawBorder` internally on `Close`.

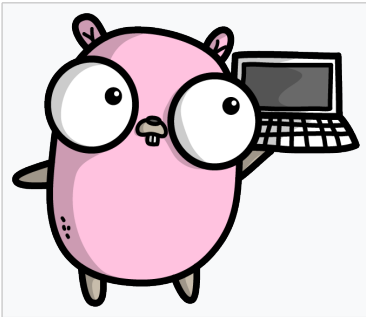
Tables

Package	Description	Category	Stars	License
gopdf	Low-level PDF generation primitives	PDF engine	3.2k	MIT
	Foundation for all Nautilus drawing	PDF engine	—	MIT
uniseg	Unicode segmentation & grapheme clusters	Unicode	800+	MIT
gomoji	Emoji detection and listing	Emoji	400+	MIT
layout	Platypus-style layout engine	Layout	internal	MIT
chart	Highcharts-style chart renderers	Charts	internal	MIT
html	Inline HTML markup parser	Text	internal	MIT
rtl	Arabic shaping and BiDi reordering	Text	internal	MIT
border	Per-side border drawing utilities	Drawing	internal	MIT
emoji	PNG emoji substitution resolver	Emoji	internal	MIT
All packages are pure Go — no CGO required.				

Table: Nautilus sub-packages. Rows with rowspan (gopdf) and colspan (summary) are demonstrated.

Images

Images are rendered with `doc.DrawImage(path, x, y, width, height)`. The method accepts PNG and JPEG files.



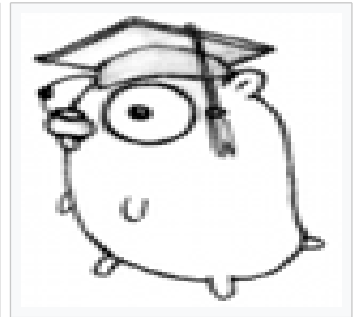
Gopher with laptop



Running Gopher

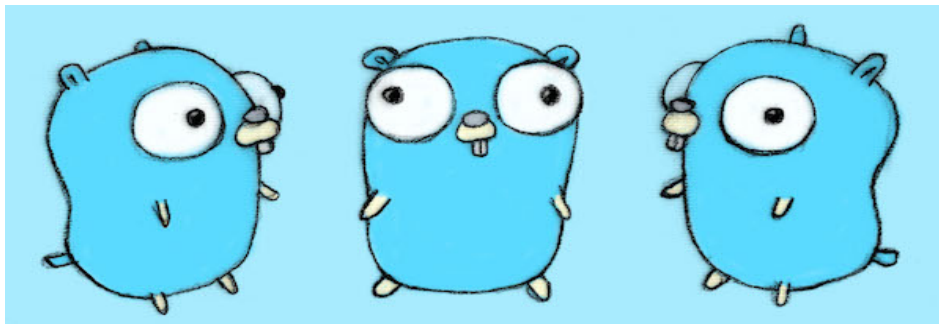


Gopher talks



Gopher with docs

Larger image — go-header.jpg:



The Go Gopher was designed by Renee French and first appeared in 2009 when Go was first announced. The Gopher has become the beloved mascot of the Go community, appearing in countless community contributions, conference talks, and merchandise.

Layout Engine — DocTemplate / PageTemplate / LayoutFrame / Flowables

The Layout Engine

Go is an open source programming language that makes it simple to build secure, scalable software. Developed at Google, Go combines the simplicity of scripting with the performance of compiled languages. The Go runtime includes garbage collection, goroutines, and channels — primitives that make concurrent programming straightforward and powerful.

Goroutines

Goroutines are lightweight threads managed by the Go runtime. Unlike OS threads, goroutines start with just a few kilobytes of stack space that can grow and shrink on demand. You can launch thousands of goroutines without exhausting system memory.

Go Modules

Go modules, introduced in Go 1.11 and made the default in Go 1.16, provide a robust dependency management system. Each module has a `go.mod` file that declares its dependencies with exact version pinning, ensuring reproducible builds across machines and time.

Static Binaries

The Go compiler produces statically linked native binaries by default. This means your entire application — including all its dependencies — is bundled into a single executable file, simplifying deployment dramatically: no virtual environments, no shared libraries to manage.

Standard Library

Go's standard library is comprehensive: it includes packages for HTTP, cryptography, JSON, SQL, testing, and much more. The toolchain ships as a single binary and supports cross-compilation to dozens of operating system and architecture combinations with a single command.

The Gopher Mascot

The Go Gopher was designed by Renee French and first appeared in 2009 when Go was first announced. The Gopher has become the beloved mascot of the Go community, appearing in countless community contributions, conference talks, and merchandise.

Nautilus makes PDF generation in Go straightforward, composable, and beautiful.

Wrap is always called before Draw or Split. SpaceBefore is suppressed at the top of a frame (atTop flag). KeepTogether wraps a group and issues a FrameBreak if the group does not fit. CondPageBreak checks remaining frame height and forces a page break when below the threshold.

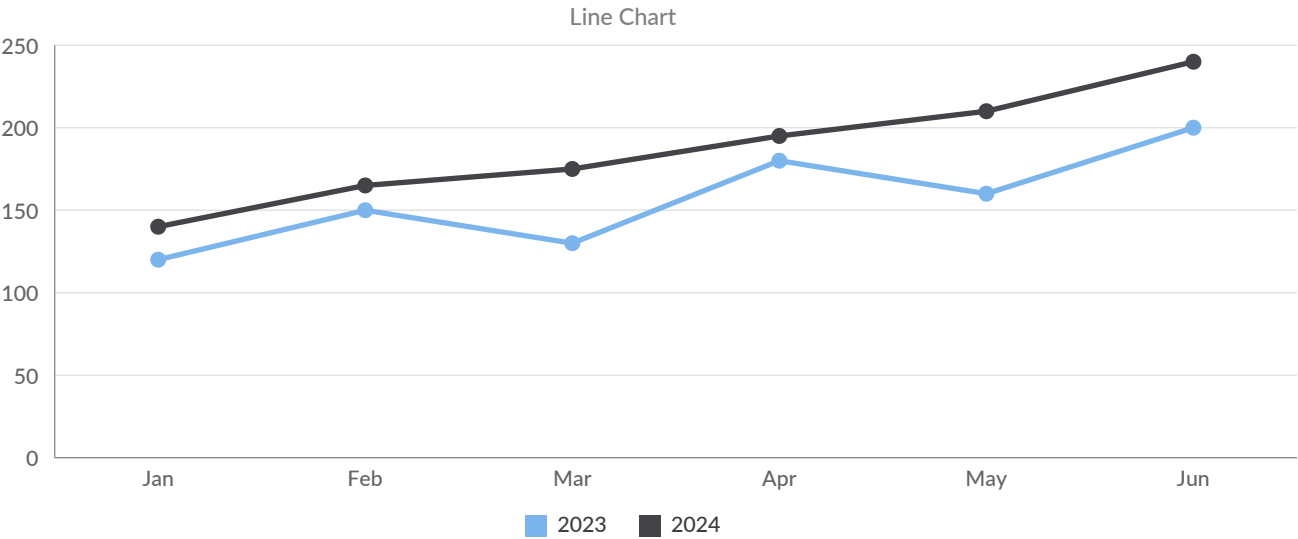
More About Go

Go's concurrency model is based on communicating sequential processes (CSP). Rather than sharing memory between goroutines and protecting it with locks, Go encourages passing data through channels. This design leads to programs that are easier to reason about and less prone to data races.

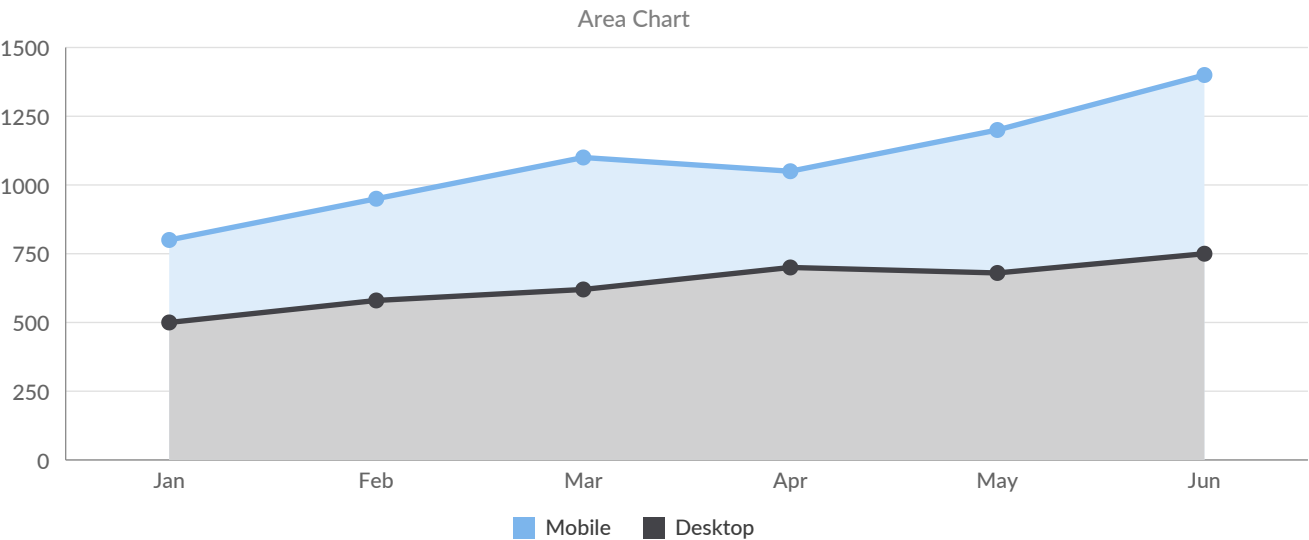
The Go toolchain includes `go build`, `go test`, `go vet`, `go mod`, and `go generate` among others. The ecosystem has grown to include powerful tools like `golangci-lint`, `staticcheck`, and `gopls` (the official language server) that integrate seamlessly with editors.

Performance benchmarks consistently show Go programs executing within 2x of equivalent C++ programs while offering dramatically higher programmer productivity. This balance makes Go an excellent choice for network services, CLI tools, and system software.

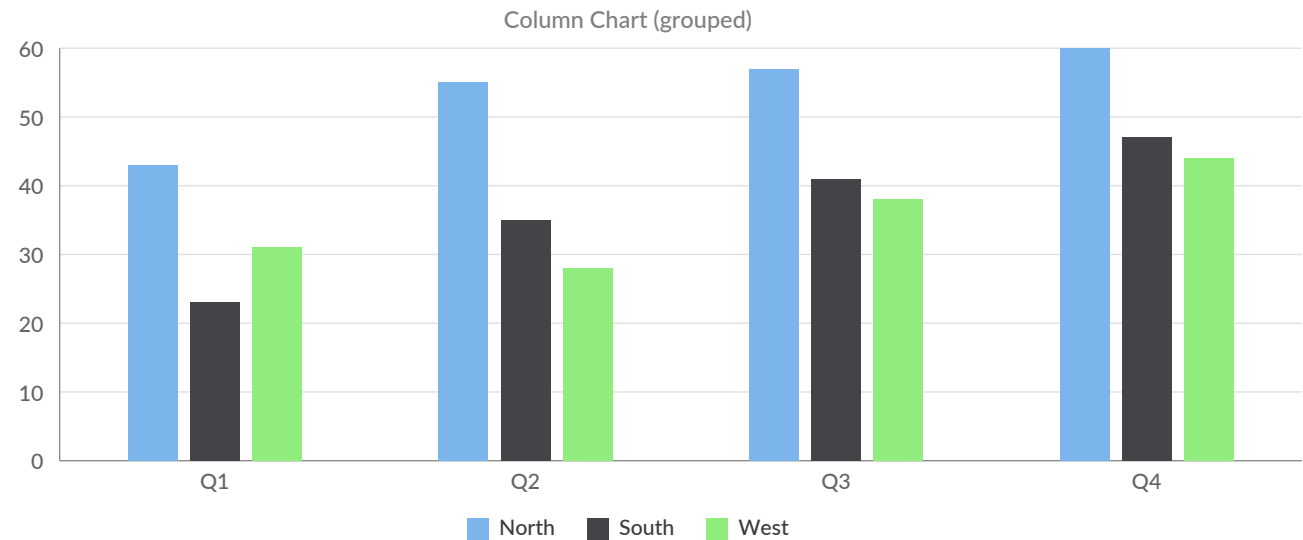
Monthly Revenue 2023 vs 2024



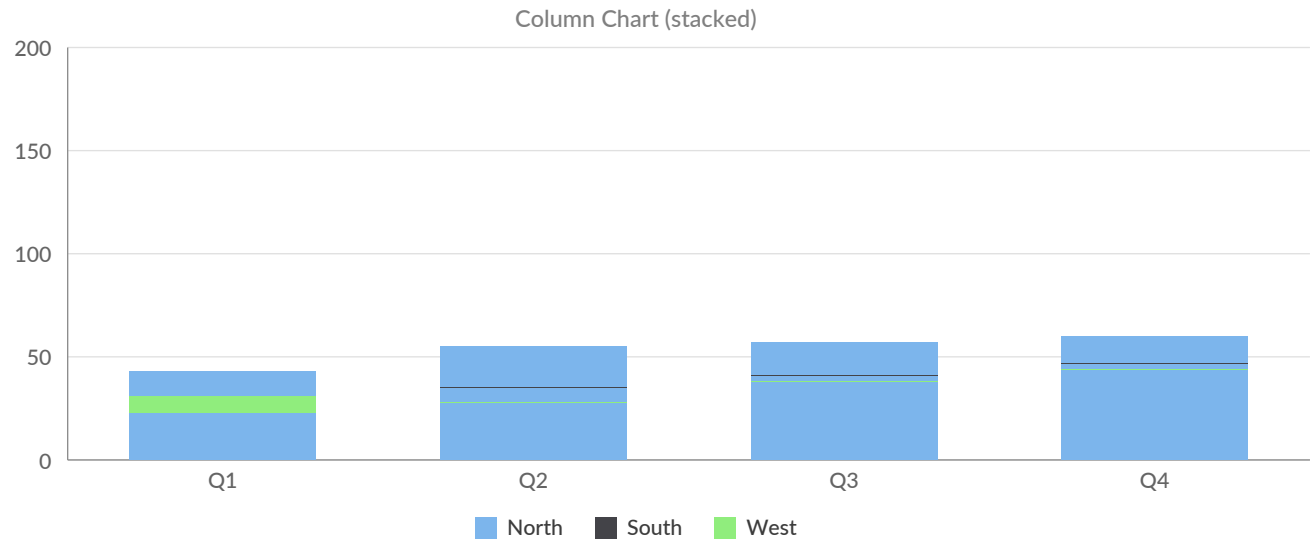
Website Visitors by Platform



Quarterly Sales by Region

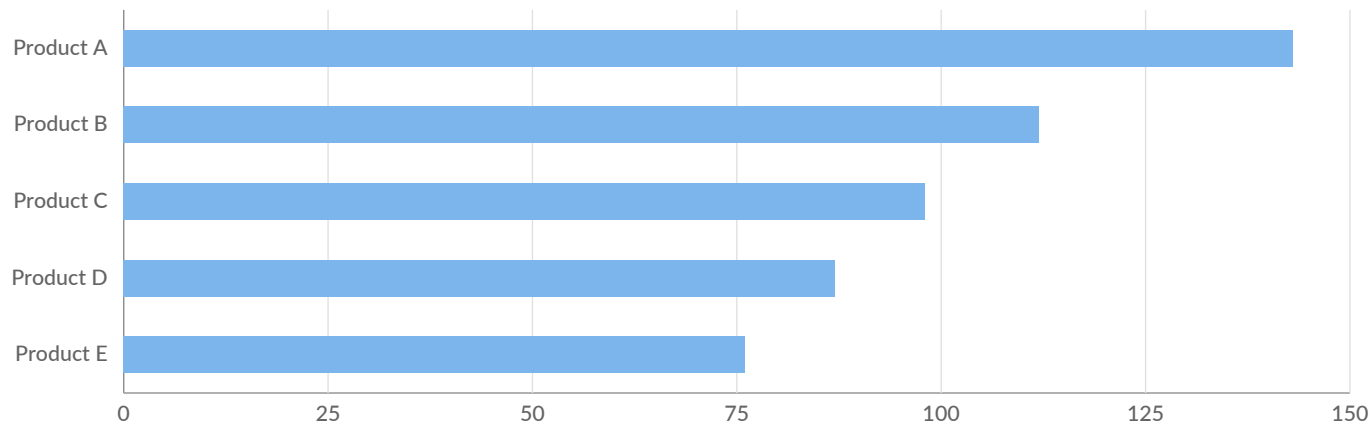


Stacked Regional Sales



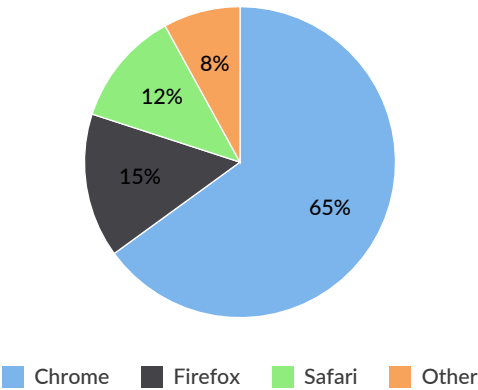
Top 5 Products by Units Sold

Bar Chart (horizontal)



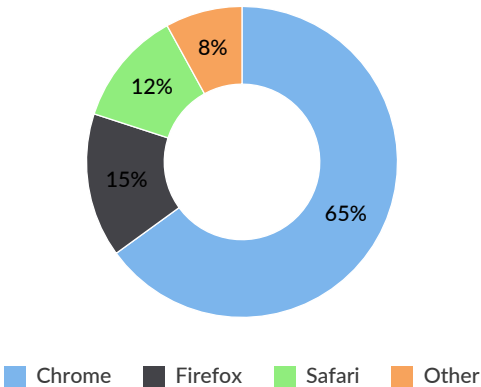
Browser Market Share

Pie Chart



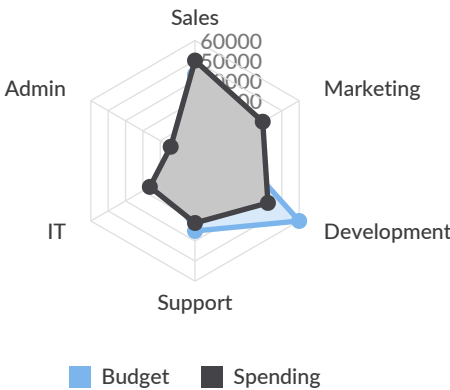
Browser Share (Donut)

Pie Chart with InnerSize=50%

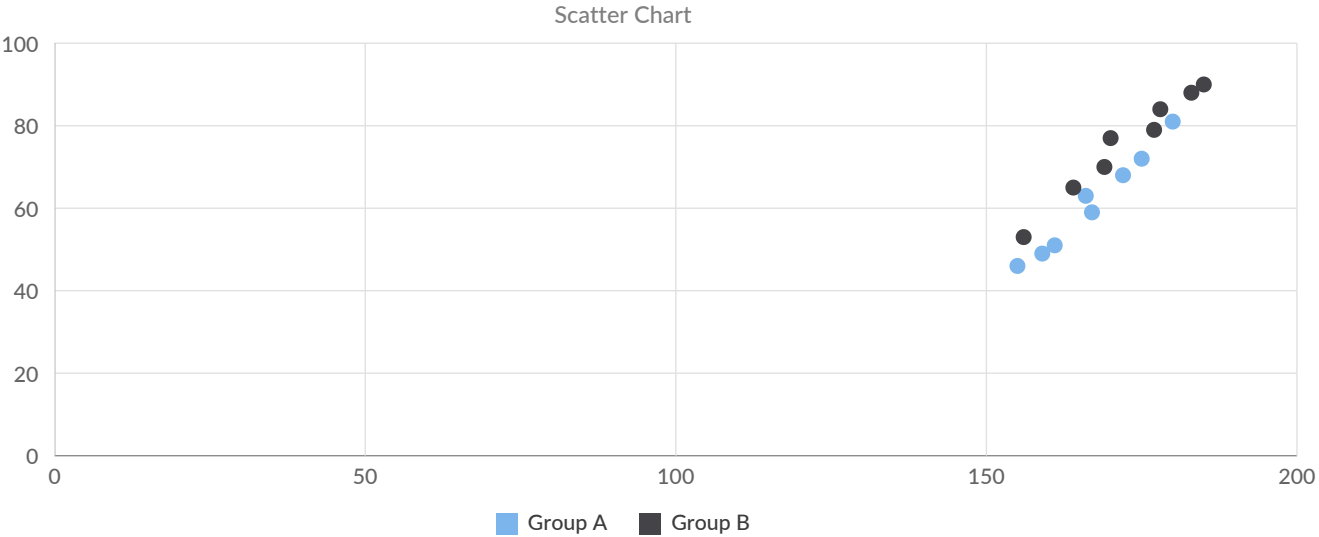


Budget vs Spending

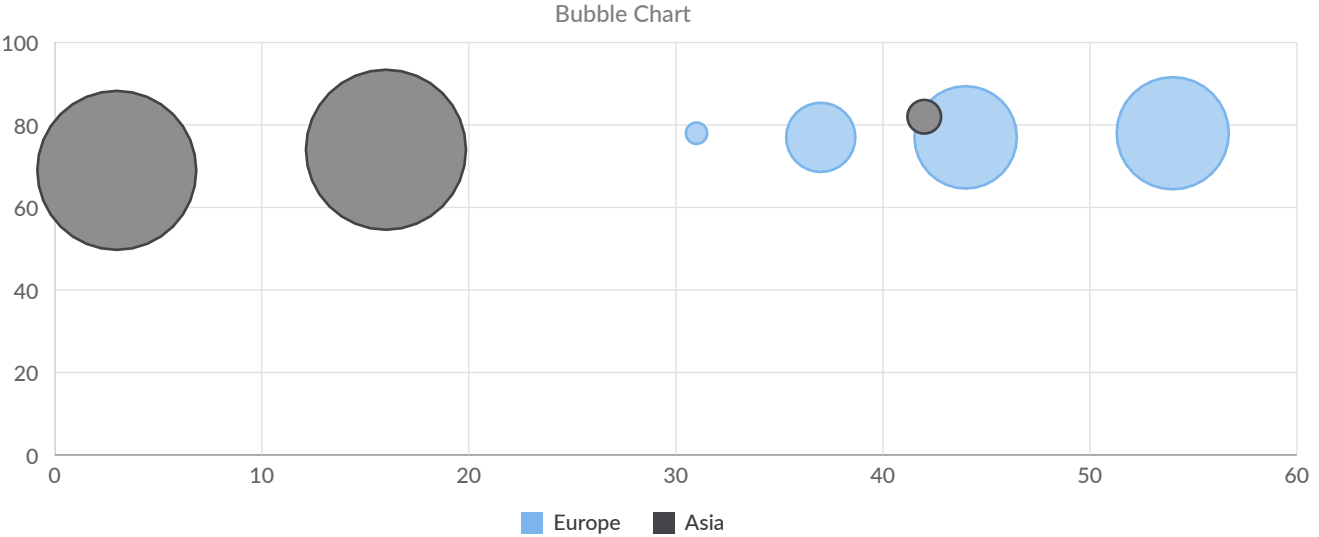
Polar / Spider / Radar Chart



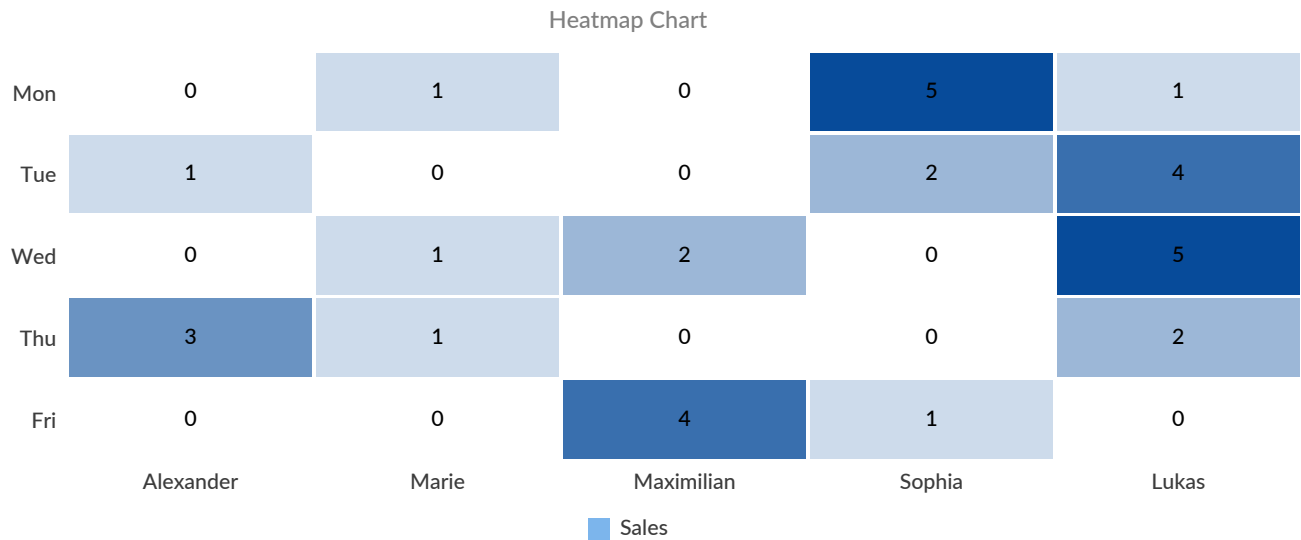
Height vs Weight Scatter



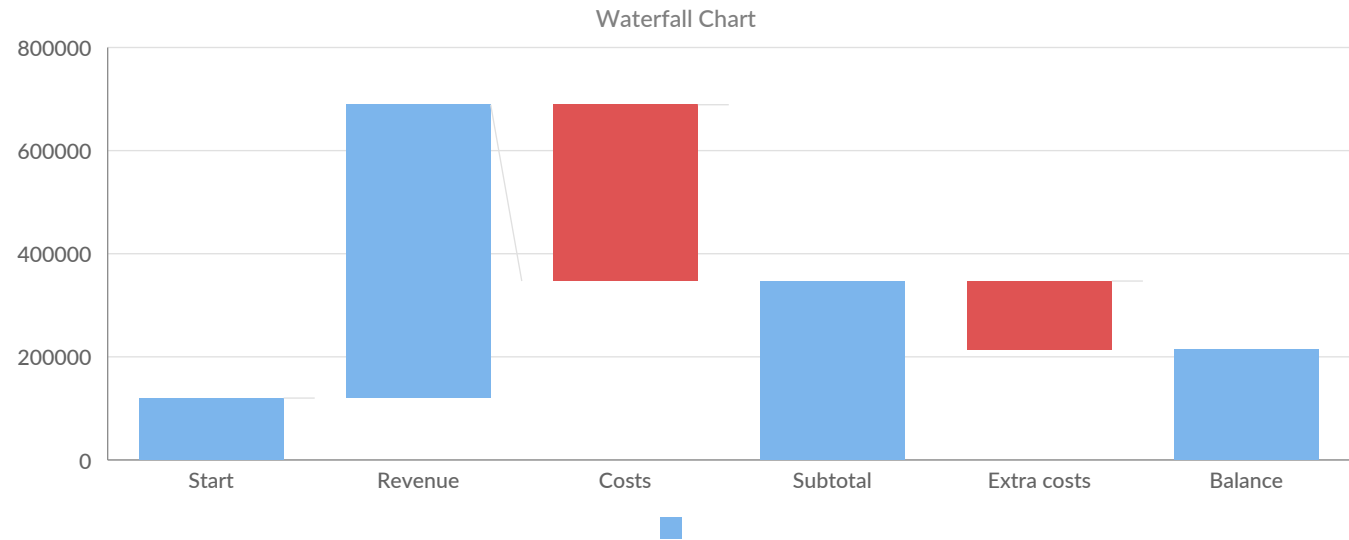
GDP, Life Expectancy, Population



Sales per Employee per Weekday

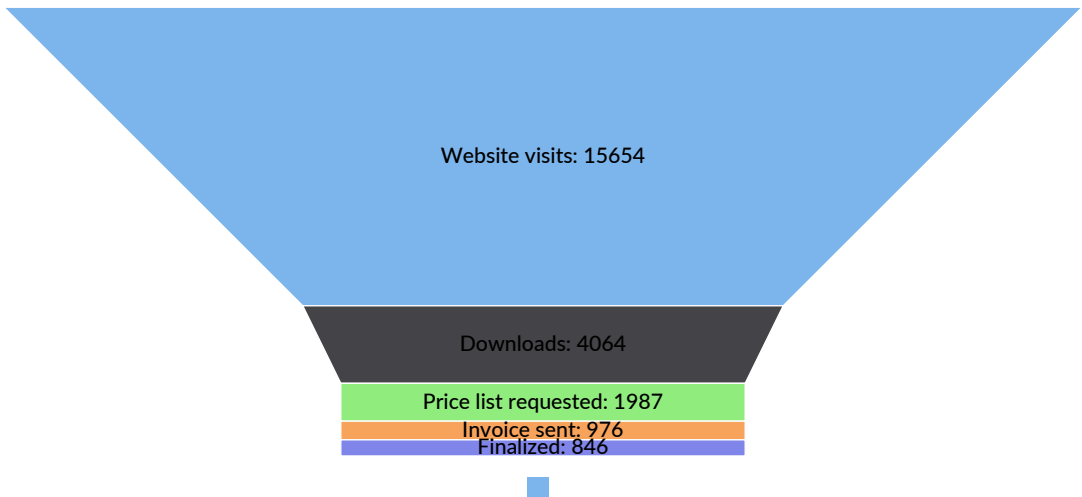


Company Financials Waterfall



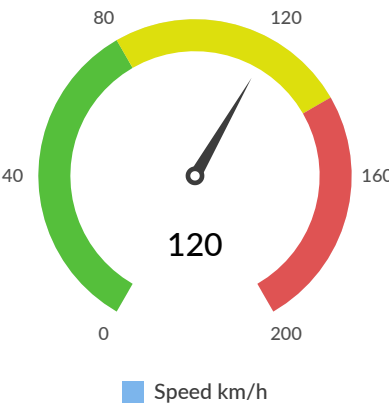
Sales Funnel

Funnel Chart



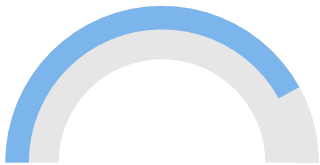
Speedometer

Gauge Chart



Activity Level – 84%

Solid Gauge Chart

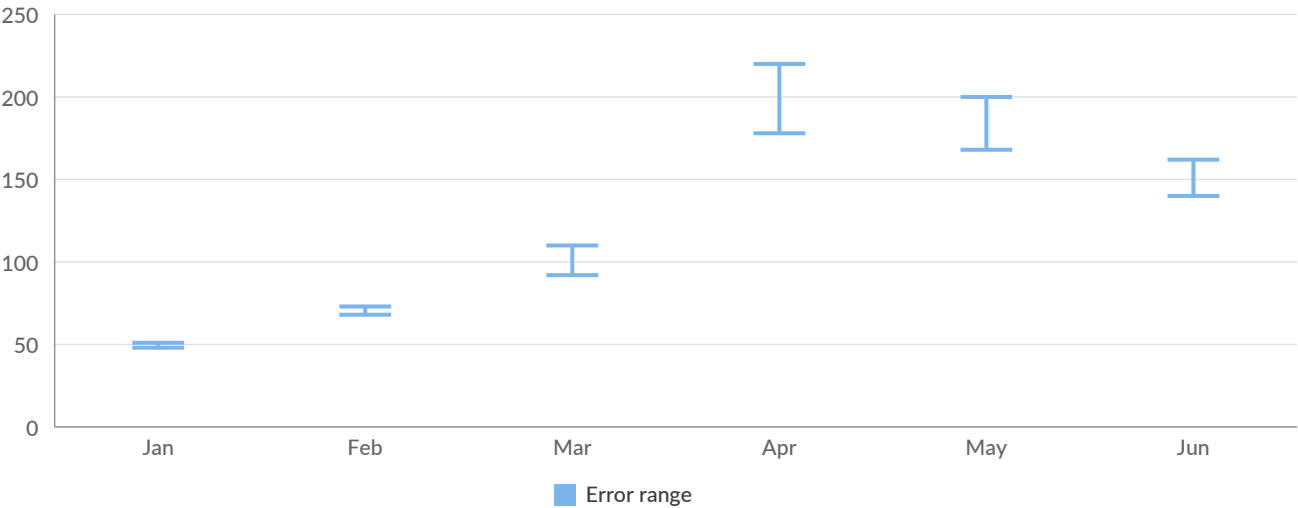


84

Activity

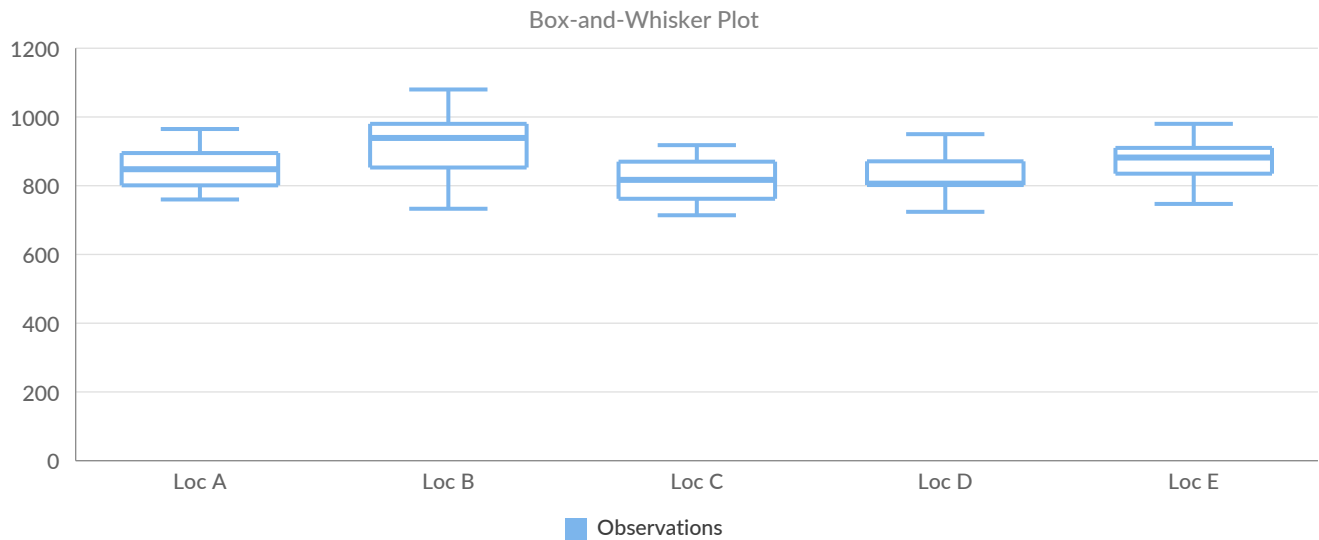
Rainfall with Error Bars

Error Bar Chart

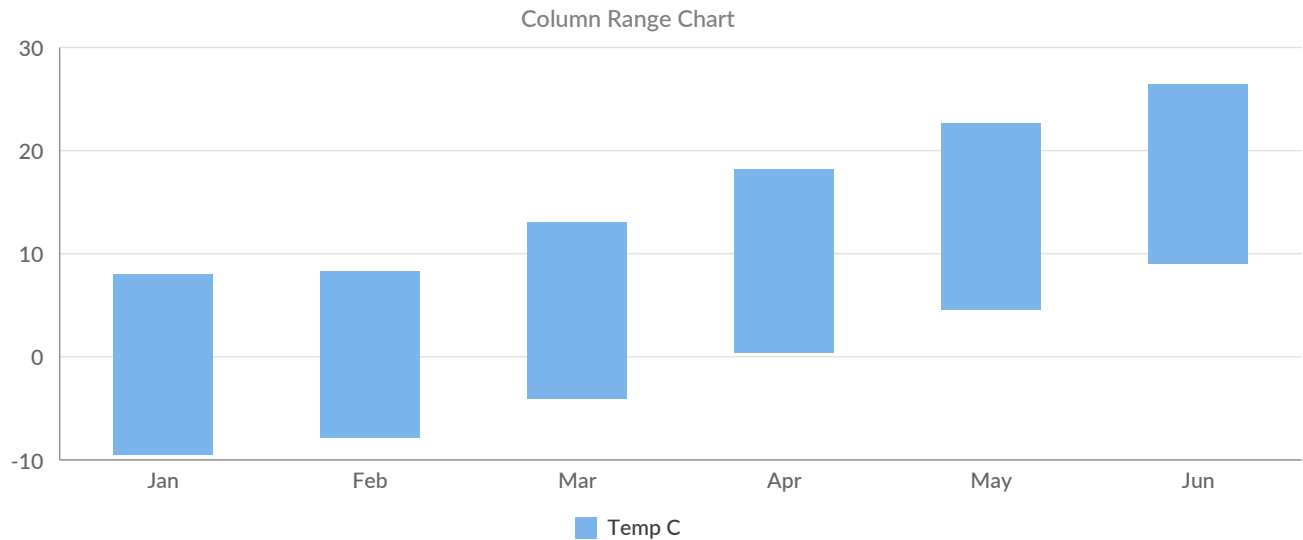


Error range

Observation Distributions

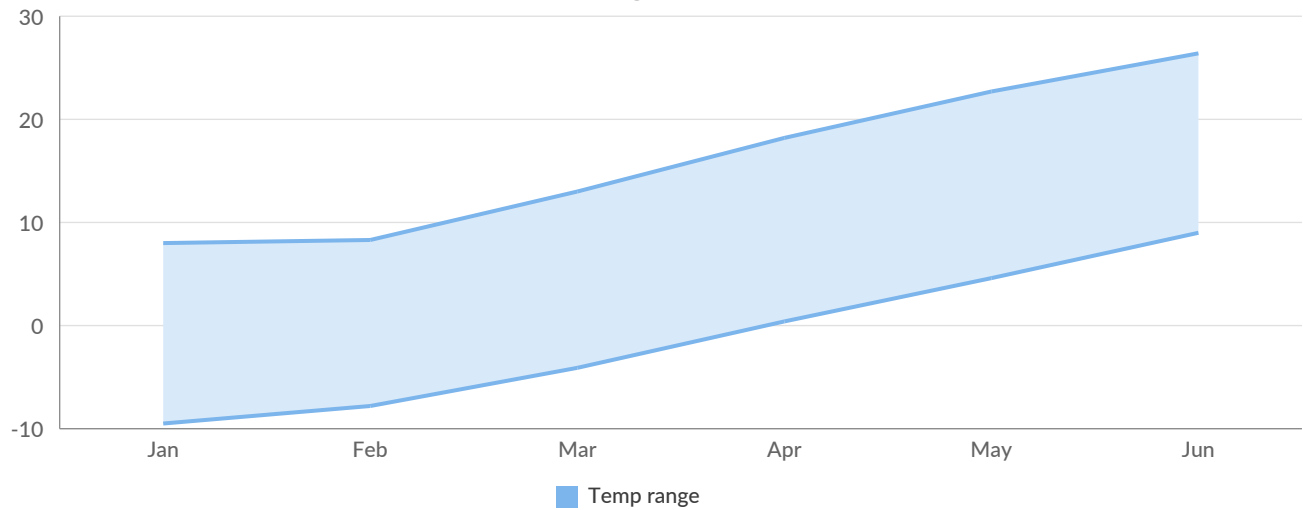


Temperature Range per Month



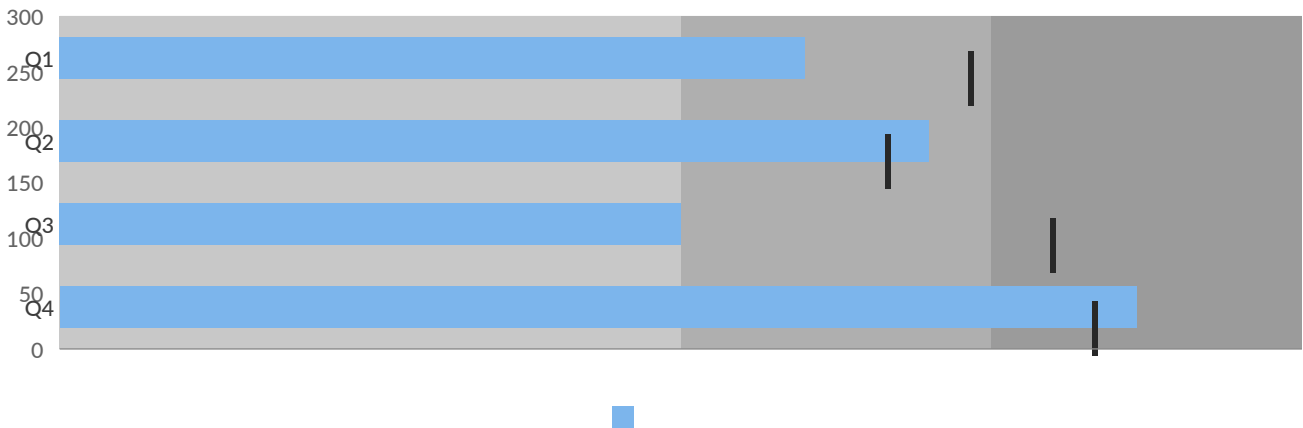
Temperature Band

Area Range Chart

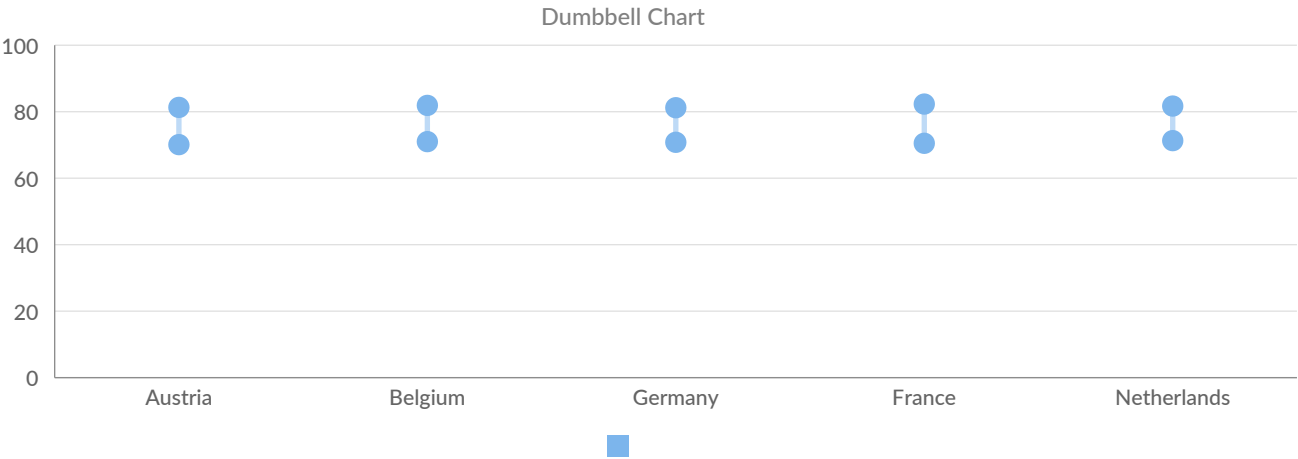


Quarterly Performance vs Target

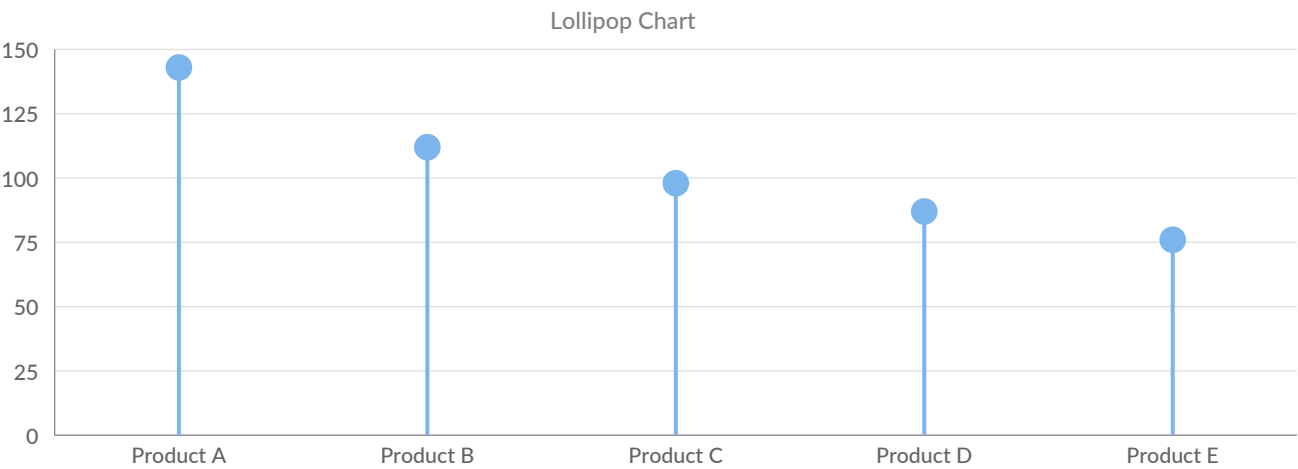
Bullet Chart



Life Expectancy 1990 vs 2020



Top 5 Products – Lollipop



Global Revenue by Region

Treemap Chart

